

quisitions are Sterling Infotech which has acquired Finnish turbine maker WinWind Oy, and ethanol technology supplier Praj Industries which bought US engineering firm CJ Schneider.

With technology getting better, the per unit capital cost of green electricity has come down drastically. As a result, wind power is at par with conventional energy in some markets. Tanti reckons that many of his early clients such as Bajaj Auto, Ajanta Clocks and Madras

picked up stakes in the US-based Stion Corporation and SolFocus, and Slovenia-based Solarvalue, had invested Rs 260 crore in setting up a photovoltaic (pv) business to manufacture solar cells and modules. The hi-tech pv cells, which convert sunlight into power, can provide energy at low cost for years with minimal maintenance. Similarly, this June, Silicon Valley-based Signet Solar announced a \$2-billion (Rs 8,000-crore) investment to set up three solar

film modules, which are 25-30 per cent cheaper than the older crystalline silicon modules.

More than cost-effectiveness, the potential of powering 80,000-odd villages in India with solar stations, compared to taking the grid route, is huge. For instance, a solar power station of 100 KW (lighting 200 houses for four-five hours a day) can be set up at a cost of \$8,00,000 (Rs 3.2 crore), while it takes \$8,000-10,000 (Rs 3,20,000-

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4,00,000) per km for cabling besides the costs of installation, distribution and generation. Hundreds of kilometres need to be covered. In what could be a big boost to solar power, bids have been invited for the Government rural electrification programmes in Madhya Pradesh, Uttar Pradesh, Bihar and Leh.

A report by brokerage firm CLSA says the pv sector is set to grow at 38 per cent year-on-year till 2010 to a total capacity of 6 GW, taking the industry turnover to \$35-40 billion (Rs 1,40,000 crore-Rs 1,60,000 crore). But for that to happen, the technology has to compete with the grid—it is three times more costly at present. Currently, 60 companies are involved in the production of various pv products. These include Tata BP Solar, which is investing \$100 million (Rs 400 crore) to create a solar plant in Bangalore. With the new

semiconductor policy that offers a host of incentives, solar cells can attract multibillion dollar investments.

There has been less headway in bio-diesel and bio-fuel despite the potential. There are about a dozen small-scale bio-diesel projects across India. Among the more ambitious projects are Natural Bioenergy's 300-tonne export-oriented facility in Andhra Pradesh and the upcoming Jain Irrigation, which plans to set up a 1,50,000-tonne bio-diesel plant in

INVESTMENTS IN ALTERNATIVE ENERGY COMPANIES

COMPANY	BUSINESS	INVESTOR	YEAR	AMOUNT
Suzlon	Wind turbine manufacturing	■ Citigroup Venture Capital International	2006	100
		■ ChrysCapital		100
Praj Industries	Technology for ethanol	Vinod Khosla	2006	117
Shriram EPC	Biomass power plants, windmills	Bessemer Venture Partners	2006	100
Reva	Making electric cars	■ Draper Fisher Jurvetson	2006	36
		■ Global Environment Fund		36
		■ Fursa Alternative Strategies**		8
WinWind Oy (Finland)	Wind energy	Sterling Infotech Group	2006	125
NEPC	Wind energy	■ Reliance Capital	2006	96
		■ Nimesh Shah		15
		■ Tarun Jain		7.2
Natural Bio Energy	Bio-diesel—jatropha	■ APIDC Capital ■ SIDBI VC ■ UTI VC	2005	—
Moser Baer	Making solar cells, modules	Warburg Pincus	2004	685*

* Total investment in the company. ** Formerly HBV Mellon. Amount in Rs crores

Cement are now enjoying power at a fixed cost of 50 paise per unit which will continue till the life of the windmill. Today, wind power installations in India are growing at 48 per cent. In 2005, the country surpassed Denmark to bag the No. 4 place in installed wind power capacity after Germany, Spain and the US. Says Prabhu Nanjani, V-P, Essar Power: "There is a worldwide market. The more you produce, the more you sell."

The market for solar energy is as hot. In 2005, Moser Baer, which has

pv making plants in India, each with an annual capacity of 300 MW.

The opportunity is huge. The global pv cell market stands at \$7-8 billion (Rs 28,000-32,000 crore) and is expected to grow to \$40 billion (Rs 1,60,000 crore) by 2010. Much of the production growth could come from India and China. "Around 136 countries are in the process of enacting pro-solar legislations. This will trigger growth," says Ratul Puri, executive director, Moser Baer. He is betting on thin



"A long-term purchase policy for utilities must be made mandatory."

TULSI TANTI, CMD, SUZLON

THE WIND TURBINE maker has orders of 2,880 MW including exports.